

Wahid Mahmood

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[Google Scholar](#)

Research Interests

Computer Vision, Explainable AI, Agentic AI, Explainable AI.

Education

Khulna University of Engineering & Technology (KUET)

Jan 2019 – Feb 2024

B.Sc. in Computer Science and Engineering (CSE)

- **GPA:** 3.53/4.0 ([Transcript](#))

- **Undergraduate Thesis:** *Study on Agricultural Land Reduction in Bangladesh using Satellite Images.*

Supervisor: Dr. Sk. Md. Masudul Ahsan.

1. Built a Sentinel-2 change detection dataset of 500+ image pairs and benchmarked it against deep learning baselines
2. The work produced two conference papers.

- **Relevant Coursework:** Natural Language Processing, Data Mining, Image Processing and Computer Vision, Linear Algebra, Statistics.

Publications

W. Mahmood, T. Joty & S. M. M. Ahsan, "NSNUNet: A Neighbourhood Attention Based Siamese Nested UNet for Change Detection using Satellite Images."

Dec 2024

IEEE International Conference on Computer and Information Technology (ICCIT), 2024. DOI:

[10.1109/ICCIT64611.2024.11022454](https://doi.org/10.1109/ICCIT64611.2024.11022454)

- Proposed a neighbourhood attention-based Siamese nested UNet for remote sensing change detection.
- Achieved 88.43% F1 (+1.25 points) on LEVIR-CD and 92.14% F1 (+0.21 points) on the Agricultural Land dataset (ours) over baseline SNUNet.
- Outperformed established baselines (SNUNet, IFN, FC-Siam-diff, FC-Siam-conc) across both benchmarks.

W. Mahmood & S. M. M. Ahsan, "A Semi-Supervised Approach for Detecting Agricultural Land Reduction in Bangladesh Using Satellite Images."

Sep 2024

IEEE International Conference on Computing, Communication, and Intelligent Systems (COMPAS), 2024. DOI:

[10.1109/COMPAS60761.2024.10795941](https://doi.org/10.1109/COMPAS60761.2024.10795941)

- Developed a semi-supervised pipeline combining machine learning (XGBoost) with iterative morphological refinement, eliminating the bottleneck of manual pixel-level annotation for multispectral Sentinel-2 imagery.
- Created a multi-temporal remote sensing benchmark of 500+ image pairs over Bangladesh, validating deep learning architectures to quantify an 8.35% (19.86 km²) reduction in regional agricultural land between 2018 and 2022.

Manuscripts in Preparation

W. Mahmood, N. Hasan, M. Mahadi, S. Shuvo, T. Joty & N. Siddique, "Unveiling Overfitting Patterns: Explainable AI-Driven Benchmarking of LLMs, Deep Learning Models, and a Hybrid Framework for Medical Image Classification."

In preparation. Collaboration with Dr. Nazmul Siddique, Ulster University.

- Formulated a methodology to compare interpretability and find out if deep learning architectures and vision-language models are properly highlighting the correct features from images by reflecting their generalisations, or whether they are overfitting or hallucinating.
- Designed the evaluation pipeline end-to-end: image preprocessing, dataset curation, and a hybrid deep learning model built as a comparative baseline.

Experience

Machine Learning Engineer, ReliSource Technologies Ltd. – Dhaka, Bangladesh

Jun 2024 – Present

- **Protocol Extractor.**

1. Architected an agentic Retrieval-Augmented Generation (RAG) pipeline to automate information extraction from complex clinical trial protocols.
2. Integrated intelligent markdown header-based chunking, cross-encoder reranking, and other methodologies to optimize accuracy up to 85% on BERTScore and semantic similarity metrics.
3. Included automated response validation and a Human-in-the-Loop (HITL) framework for gradual improvement using few-shot learning.
4. Full offline inference backend orchestrated via LangGraph, FastAPI, LM Studio, and Qdrant.

- **Agentic Medical Inventory Manager:**

1. Built an end-to-end forecasting and restocking platform that pairs time-series forecasting with an agentic LLM decision loop.
2. Pulled news, weather, and holiday data to compute contextual demand multipliers and used integer Linear Programming to generate budget-constrained restocking recommendations.

- **AI Chatbot:** RAG-based assistant with source citations and streamed responses, tuned to a 4-second end-to-end response time (LangChain, FastAPI, OpenAI).

- **Drug Discovery Pipeline:** Built an ML pipeline to screen and validate candidate compounds, including ADMET property analysis, using AlphaFold, DeepChem, Biopython, and RDKit.

- **Data Engineering:** Completed the company Azure Data Engineering programme and supported client data workflows using Azure Data Factory and Databricks.

Projects

Intent Based Product Search

Mar 2025

- Engineered an intent-aware product retrieval system for e-commerce on a fine-tuned Llama 3.2 1B model.
- Designed a hybrid search architecture with sub-2000 ms query latency, and instrumented it with LangSmith tracing to track retrieval and answer quality.
- **Tools:** LangChain, FastAPI, LangSmith, PGVector, Ollama, Docker, Redis.

Open-CD (open-source contribution)

Feb 2024

- Modified version of the open-source change detection framework named Open-CD.
- Added the feature of loading images with more than 3 channels and running all the models on a specific dataset.
- **Tools:** Python, OpenMM, Rasterio.

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July 2023

- E-commerce app connecting users with nearby stores, with shop location tracking.
- **Tools:** Flutter, Firebase.

Skills

Programming: Python, C/C++, Java, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, OpenCV, statsmodels

LLM & Agentic Frameworks: LangChain, LangGraph, CrewAI, LangSmith, Langfuse, Ollama, LM Studio

Data & Vector Databases: PostgreSQL, PGVector, Qdrant, Neo4j, Redis, Snowflake, Databricks

Other Tools: Docker, Git/GitLab, MLflow, FastAPI, React, Flutter, Azure Data Factory

Standardized Test Scores

International English Language Testing System (IELTS): 7.5 (overall)

Aug 2025

Achievements

Dean's Award, KUET	2019 – 2020
• Awarded for academic excellence with a yearly GPA of 3.83 (award threshold: 3.75).	
5th place, Poridhi AI Engineering Hackathon	2025
• Placed in the top 10 out of 80 competing teams.	
7th place, Intra-KUET Programming Contest	2022
• Placed 7th out of 85 competing teams.	
Competitive Programming	2019 – 2023
• Codeforces: Specialist (max rating 1493, 650+ problems solved). • CodeChef: 4 star (max rating 1864).	

References

Dr. Sk. Md. Masudul Ahsan	Thesis Supervisor
<i>Professor, Department of Computer Science & Engineering</i> Khulna University of Engineering & Technology (KUET) Supervised my undergraduate thesis and co-authored both published papers, 2023–2024. Email: smahsan@cse.kuet.ac.bd	
Dr. Nazmul Siddique	Research Collaborator
<i>Senior Lecturer, School of Computing, Engineering and Intelligent Systems</i> Ulster University, United Kingdom Co-author on the explainable AI benchmarking manuscript in preparation. Email: nh.siddique@ulster.ac.uk	
Abdullah Muhammad Moosa	Industry Supervisor
<i>Functional Lead, Machine Learning</i> ReliSource Technologies Ltd. Direct supervisor at ReliSource since 2024. Email: abdullahmoosa128@gmail.com	